# Topic 7 — Electronics, Semiconductors and Computer Technology

**MANDATORY SYLLABUS SCOPE:** Covers all dimensions of Electrical Conductivity & Free Electron Theory, Conductors, Insulators, and Semiconductors, Intrinsic vs Extrinsic Semiconductors (P-type vs N-type), Doping & Carrier Generation, Diodes (P-N, Zener, Tunnel Diode, LED, Photodiode), Transistors (BJT, FET) & Integrated Circuits, Superconductivity (Heike Kamerlingh Onnes, Meissner Effect, Critical Temperature, High-Pressure Hydrides $LaH_{10}$, Maglev), Computer History & Generations (HEC-2M, TIFRAC, Generations I to V), Supercomputers (National Supercomputing Mission, PARAM Series, AIRAWAT, World & India Top500 Rankings), Computer Architecture (CPU, ALU, Control Unit, Ports, Buses), Memory Hierarchy (Registers, Cache, RAM, ROM/BIOS/UEFI, Hard Disk, DVD, Blu-ray, Flash Memory), Number Systems & Logic (Binary Arithmetic, Self-Complementing Codes, Boolean Logic), Computer Networking & Topologies (LAN, MAN, WAN, OSI 7-Layer Model, TCP/IP, IP Addressing IPv4 vs IPv6, MAC Address $48$ bits, DNS), Wireless & Optical Telecom (Wi-Fi, Bluetooth $2.4$ GHz, Li-Fi $10$ Gbps, NFC $4$ cm, Short Range Devices), Software & Programming Languages (1GL to 5GL, System vs Application, Compilers, Assemblers, SQL, Prolog), Cyber Security & Malware (Viruses, Worms, Trojans, Ransomware/WannaCry, Spyware/Pegasus, Stuxnet, Firewalls, DoS/DDoS, Encryption/Blowfish, PKI), Artificial Intelligence & Frontier Tech (John McCarthy, Turing Test, ML, Deep Learning, Generative AI / ChatGPT, FPGA, Quantum Qubits & Majorana 1, Paris AI Action Summit 2025), and Digital India & E-Governance (4 Pillars, Gartner Model, IT Act 2000, BHIM UPI, Aadhaar, DigiLocker, CoWIN, Web 3.0, 3D Printing). Fully synchronized with Ghatna Chakra Physics and Computer & IT, NCERT Class 11–12, and latest 2021–2025 State PCS PSC papers.

## Current Affairs (2024–2026 Focus)

1. **AI Action Summit, Paris (February 2025 - IAS Pre 2025 / 71st BPSC Fact):** - Held at the Grand Palais, Paris on 10–11 February 2025, co-chaired by **Prime Minister Narendra Modi (India)** and **President Emmanuel Macron (France)**. - Built upon the Bletchley Park AI Safety Summit (UK, 2023) and the Seoul AI Summit (South Korea, 2024). - Over 50 nations (including India, France, and China) signed the *Statement on Inclusive and Sustainable AI for People and the Planet*. - **Common Trap Fact:** The **USA and the UK DID NOT sign** this declaration!

2. **Quantum Computing Processors: Majorana 1 & Ocelot (IAS Pre 2025 Fact):** - **Majorana 1:** Unveiled by **Microsoft** on 19 February 2025, utilizing topological qubits aimed at achieving a million-qubit scale. - **Ocelot:** Quantum processor developed by **Amazon Web Services (AWS)**. - Theoretical subset fact: **Deep Learning is a subset of Machine Learning**, which is in turn a subset of Artificial Intelligence ($AI \supset ML \supset DL$).

3. **National Supercomputing Mission (NSM) Milestones (November 2025 Status):** - Launched in 2015, jointly steered by **Department of Science and Technology (DST)** and **Ministry of Electronics and Information Technology (MeitY)**; implemented by **C-DAC (Pune)** and **IISc (Bengaluru)**. - Deployed **37 supercomputers** with a cumulative capacity of $39$ Petaflops, connected over the **National**

**Knowledge Network (NKN)** with an overall utilization rate exceeding $85\%$. Five training centres established at Pune, Kharagpur, Chennai, Palakkad, and Goa. - **Fastest Supercomputer Installed in India: PowerEdge XE 9680** (installed at Shakti Cloud, Yotta Data Services, Mumbai; ranked 28th globally; $R_{\max} = 84.31\ \text{Petaflops}$, manufactured by Dell, USA). - **Fastest Indigenous Supercomputer Developed in India: AIRAWAT-PSAI** (installed at C-DAC Pune; ranked 188th globally; $R_{\max} = 8.50\ \text{Petaflops}$, $R_{\text{peak}} = 13.17\ \text{Petaflops}$). - **World's Top 3 Supercomputers (Nov 2025):** 1. **El Capitan (USA)** ($1.809\ \text{Exaflops}$), 2. **Frontier (USA)**, 3. **Aurora (USA)**.

4. **PARAM Series Deployment in India:** - **PARAM 8000:** India's first indigenous supercomputer, built by C-DAC Pune in 1991 under **Dr. Vijay Bhatkar** ("Father of Indian Supercomputers"). - Key institutional deployments:
   - **PARAM Kamrupa:** IIT-Guwahati (inaugurated by President Droupadi Murmu, Oct 2022; $838\ \text{TFLOPS}$).
   - **PARAM Shivay:** IIT-BHU (first installed under NSM).
   - **PARAM Shakti:** IIT-Kharagpur.
   - **PARAM Pravega:** IISc Bengaluru ($3.3\ \text{Petaflops}$).
   - **PARAM Ganga:** IIT-Roorkee.
   - **PARAM Ananta:** IIT-Gandhinagar.
   - **PARAM Sanganak:** IIT-Kanpur.
   - **PARAM Rudra:** High-performance systems at IUAC Delhi, IIT-Bombay, IIT-Madras, NCRA Pune, SNBNCBS Kolkata, IIT-Patna, and C-DAC Delhi.
   - **ANUPAM Series:** Supercomputer series developed indigenously by **Bhabha Atomic Research Centre (BARC)** for internal nuclear simulations (first was Anupam 860/4 in 1991).
5. **Historical Firsts in Indian Computing:** - **HEC-2M (Hollerith Electronic Computer-2M):** India's first digital computer, imported from England (designed by A.D. Booth) and installed in **1956 at the Indian Statistical Institute (ISI) in Calcutta**. - **TIFRAC (TIFR Automatic Calculator):** First digital computer built in India, commissioned in **1960 at TIFR Mumbai**.
6. **E-Governance Initiatives (2024–2025 Focus):** - **Suvidha 2.0 App:** Upgraded mobile application launched by the **Election Commission of India (ECI)** in October 2024 as a one-stop portal for candidates and political parties to track nominations, schedules, and campaign permissions. - **Gartner's 4-Phase Evolution Model of E-Governance (MPPSC Pre 2025 Fact):**

$$\textbf{Information} \rightarrow \textbf{Interaction} \rightarrow \textbf{Transaction} \rightarrow \textbf{Transformation}$$

   - **Four Pillars of E-Governance: People, Process, Technology, Resources** (MPPSC Pre 2024).
7. **Generative AI & Enterprise Breakthroughs:** - **ChatGPT:** Developed by OpenAI, launched on **November 30, 2022**, based on the Generative Pre-trained Transformer Large Language Model architecture. - **Maharaja:** World's first airline Generative AI virtual agent deployed by **Air India** (2023), powered by Microsoft's Azure OpenAI service, handling over 1,300 customer query domains. - **FPGA (Field Programmable Gate Array):** Reconfigurable semiconductor ICs widely deployed for hardware acceleration in AI inference and edge computing (MPPSC Pre 2025).

## Detailed In-Depth Core Theory

### 7.0 Semiconductor Physics, Electrical Conductivity & Solid-State Electronics

1. **Conductors, Insulators and the Conductivity Hierarchy:** - **Conductors:** Contain vast concentrations of free conduction electrons.
   - **Electrical Conductivity Ranking: Silver ($Ag$) is the BEST conductor of electricity**, followed in order by **Copper ($Cu$)**, **Gold ($Au$)**, and **Aluminium ($Al$)**.
   - Copper is widely used for residential and industrial wiring due to high conductivity and cost-effectiveness compared to silver.
   - **Mica:** A mineral with unique physical properties—it is an excellent **thermal conductor** but a powerful **electrical insulator** (used in electric irons and heating elements).

2. **Semiconductors & Temperature Behavior:** - Materials having electrical resistivity intermediate between conductors ($10^{-8}\ \Omega \cdot m$) and insulators ($10^{12}\ \Omega \cdot m$). - **Temperature Coefficient of Resistance:**
   - In metals/conductors, electrical resistance increases with temperature ($\alpha > 0$) due to increased lattice phonon scattering.
   - In **Semiconductors, electrical resistance DECREASES upon heating** (conductivity increases; $\alpha < 0$) because thermal energy excites valence electrons across the narrow forbidden band gap ( $E_g \approx 1.1\ \text{eV}$ for Silicon; $0.7\ \text{eV}$ for Germanium) into the conduction band, generating electron-hole pairs.
   - At absolute zero ($0\ \text{K}$), a pure semiconductor acts as a **perfect insulator**.

3. **Intrinsic vs Extrinsic Semiconductors & Doping:** - **Intrinsic Semiconductors:** Pure elemental crystals of **Silicon ($Si$)** and **Germanium ($Ge$)** from Group 14 of the periodic table, each with 4 valence electrons. - **Extrinsic Semiconductors (Doping):** The deliberate addition of minute trace impurities to pure semiconductors to dramatically amplify their electrical conductivity:
   - **P-Type Semiconductor:** Produced by doping pure Silicon or Germanium with **Trivalent impurities (Group 13: Boron, Aluminium, Gallium, Indium)**. Creates an excess of positive electron vacancies (**holes**), which act as the **majority charge carriers**.
   - **N-Type Semiconductor:** Produced by doping pure Silicon or Germanium with **Pentavalent impurities (Group 15: Phosphorus, Arsenic, Antimony)**. Introduces extra donor electrons into the conduction band; **electrons** act as the **majority charge carriers**.

4. **Solid-State Electronic Devices:** - **P-N Junction Diode & Rectification:**
   - When P-type and N-type semiconductors are fused, a depletion region forms. A diode conducts current readily in forward bias but blocks current in reverse bias.
   - **Rectifier:** An electronic circuit device built using diodes that converts **Alternating Current (AC) into Direct Current (DC)**. (Conversely, an **Inverter** converts Direct Current into Alternating Current).
   - **Transistors:**
   - Three-terminal semiconductor devices (Emitter, Base, Collector) used for electrical amplification and high-speed digital switching. Fabricated primarily from **Silicon ($Si$)** or **Germanium ($Ge$)**. Invented in 1947 by John Bardeen, Walter Brattain, and William Shockley at Bell Labs.

- **Integrated Circuits (ICs):**
- Miniature electronic circuits comprising millions to billions of transistors, diodes, and resistors fabricated on a single tiny wafer of **Silicon ($Si$)**; invented in 1958 by **Jack Kilby** (Texas Instruments) and **Robert Noyce** (Fairchild Semiconductor).

5. **Computer Pioneers, Operating Systems & Infotech Foundations:** - **Ada Lovelace (Lady Ada Augusta Lovelace):** Celebrated as the **world's first computer programmer** for creating an algorithm for Charles Babbage's mechanical Analytical Engine to calculate Bernoulli numbers. - **Charles Babbage:** Designed the Difference Engine and Analytical Engine; known as the **Father of the Computer**. - **Linux:** An open-source, Unix-compatible operating system kernel created in **1991 by Linus Torvalds**. - **Singular Form of Data:** The technical singular form of the plural word 'data' is **Datum**. - **Junk Electronic Mail:** Unsolicited bulk messages across the Internet are classified as **Spam**. - **MICR (Magnetic Ink Character Recognition):** Magnetic character-reading technology used primarily in the **banking industry** to process bank cheques and drafts efficiently. - **Distinction Alert:** Terms like *Login, Modem, Password, Spam, LAN* belong to infotech/cyber terminology. In contrast, **Pinaka** is an indigenous multi-barrel rocket launcher (defence system), NOT an infotech term.

### 7.1 Microcomputer Architecture & Hardware Interfaces

1. **CPU Subsystems:** - **Control Unit (CU):** The central directing supervisor; fetches instructions from memory, decodes operation codes, and generates timing and control pulses for all peripherals. - **Arithmetic Logic Unit (ALU):** Carries out fundamental integer binary arithmetic (addition, subtraction) and logical decisions (AND, OR, NOT, comparison).
2. **Motherboard & Port Architecture:** - **Chipset:** Northbridge (historically connected CPU to high-speed RAM and PCIe graphics) and Southbridge (manages slower I/O peripherals, USB, SATA). Modern CPUs integrate the Northbridge directly onto the processor die. - **USB (Universal Serial Bus):** Standardized peripheral interconnect providing bidirectional data and DC power ($5\ \mathrm{V}$). - **Port Types:** Serial, Parallel (Centronics/LPT for legacy printers), VGA/HDMI/DisplayPort for video display, Ethernet RJ-45 for LAN.

### 7.2 Memory Hierarchy, Solid State Storage & Optical Media

1. **Volatile vs Non-Volatile Physics:** - Dynamic RAM requires periodic capacitive refreshing every few milliseconds to prevent charge dissipation across gate oxides. - Flash Memory (NAND) uses floating-gate transistors to trap electrons in insulated potential wells, retaining data for decades without electrical power.
2. **Optical Disc Data Packing:** - CD/DVD/Blu-ray store data as micro-indentations called **pits** separated by flat surfaces called **lands**. - Reducing laser wavelength (Red $650\ \mathrm{nm} \rightarrow$ Blue-violet $405\ \mathrm{nm}$) enables the optical pickup lens to focus on vastly smaller pit tracks, boosting storage from $4.7\ \mathrm{GB}$ (DVD) to $25\ \mathrm{GB}$ (Blu-ray).

### 7.3 Computer Networks, Switching & Internet Infrastructure

1. **Packet Switching Mechanics:** - Messages are fragmented into discrete data packets bearing destination and source IP addresses, sequence numbers, and error-checking checksums. Packets traverse diverse network routes independently and reassemble at the receiving host via TCP.

2. **Network Topologies:** - **Star Topology:** Central hub/switch connects all nodes; failure of a node does not disrupt the network, but central hub failure halts communication. - **Bus Topology:** Single shared linear backbone cable terminated at both ends; low cost but vulnerable to cable breakages. - **Ring Topology:** Unidirectional token-passing ring; failure of a single node breaks the entire loop unless a dual ring (FDDI) is used. - **Tree Topology:** Hierarchical combination of bus and star topologies; structure utilized by the global Internet. - **Mesh Topology:** Every node is interconnected with every other node, providing maximum redundancy and fault tolerance.

### 7.4 Cyber Security Threats, Malware Taxonomy & Defenses

1. **Malware Mechanisms:** - **Viruses:** Injected into executable files ( `.exe` , `.com` ); activates upon execution. - **Trojans:** Disguised as benign utilities to steal credentials via backdoor command-and-control channels. - **Keyloggers:** Capture physical hardware keystrokes to siphon banking credentials and passwords. - **E-mail Bombing:** Abusive transmission of thousands of duplicate messages to overflow mailbox quotas.
2. **Defensive Cryptography:** - **Symmetric Encryption (Blowfish, AES):** Uses identical secret key for encryption and decryption; extremely fast for bulk data. - **Asymmetric Encryption (RSA, Public Key Infrastructure - PKI):** Employs a mathematically linked key pair—a public key for encryption and a private key for decryption. Forms the basis of Digital Signatures under the IT Act, 2000.

### 7.5 Artificial Intelligence Paradigms, Emerging Tech & E-Governance

1. **Machine Learning Taxonomy:** - **Supervised Learning:** Trains on labeled input-output pairs to build regression or classification predictive models. - **Unsupervised Learning:** Discovers hidden patterns or clusters in unlabeled data (e.g. K-means clustering). - **Reinforcement Learning:** Software agents interact with an environment through trial and error, optimizing cumulative rewards.
2. **Quantum Computing Principles:** - Operates with quantum bits (**qubits**) capable of occupying coherent **superposition** states ($|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$), performing massively parallel calculations that solve intractable mathematical problems in polynomial time.
3. **E-Governance Models:** - **Gartner's 4-Phase Evolution:** 1. Presence (Information) $\rightarrow$ 2. Interaction (Intake) $\rightarrow$ 3. Transaction (Digital payments) $\rightarrow$ 4. Transformation (Unified smart delivery). - **Aadhaar Platform:** 12-digit unique ID using open APIs for biometric iris and fingerprint authentication; legally confirmed as proof of identity only (not citizenship or domicile).

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## Bilingual Terminology -- Electronics, Semiconductors and Computers

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Semiconductor** (अर्धचालक) | अर्धचालक | Conductivity between conductor and insulator; Si, Ge |
| **Diode** (डायोड) | डायोड | Two-terminal device; allows current in one direction only (rectification) |
| **Transistor** (ट्रांजिस्टर) | ट्रांजिस्टर | Three-terminal device; amplification and switching |
| **Integrated Circuit (IC)** (एकीकृत परिपथ) | एकीकृत परिपथ | Many transistors on a single chip; "chip" |

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Binary** (द्विआधारी) | द्विआधारी | Base-2 number system; 0 and 1 only; basis of computers |
| **Bit** (बिट) | बिट | Smallest unit of data; 0 or 1 |
| **Byte** (बाइट) | बाइट | 8 bits; basic addressable unit |
| **CPU** (केन्द्रीय प्रसंस्करण इकाई) | केंद्रीय प्रक्रमण एकक | Central Processing Unit; "brain" of computer |
| **RAM** (रैम) | यादृच्छिक अभिगम स्मृति | Random Access Memory; volatile; temporary storage |
| **ROM** (रोम) | केवल पठन स्मृति | Read Only Memory; non-volatile; permanent storage |
| **Internet** (इंटरनेट) | इंटरनेट | Global network of computers; WWW is a service on Internet |
| **Artificial Intelligence** (कृत्रिम बुद्धिमत्ता) | कृत्रिम बुद्धिमत्ता | Simulation of human intelligence in machines |
| **Machine Learning** (मशीन लर्निंग) | मशीन लर्निंग | AI where computers learn from data without explicit programming |
| **5G** (5जी) | पांचवीं पीढ़ी का नेटवर्क | 5th generation mobile network; speed up to 10-20 Gbps |
| **Nanotechnology** (नैनो प्रौद्योगिकी) | नैनो प्रौद्योगिकी | Technology at 1-100 nm scale; 1 nm = 10^-9 m |
| **Quantum Computing** (क्वांटम संगणना) | क्वांटम संगणना | Uses qubits (superposition/entanglement); exponentially faster for certain tasks |

## One-Liner Revision -- Electronics, Semiconductors and Computers

| # | Fast Fact |
|---|---|
| 1 | Silicon (Si) = most widely used semiconductor; Germanium (Ge) = used in early devices |
| 2 | N-type semiconductor = doped with phosphorus/arsenic (extra electrons) |
| 3 | P-type semiconductor = doped with boron/indium (holes = positive charge carriers) |
| 4 | Diode = PN junction; allows current in forward bias only; used in rectifiers, LEDs |
| 5 | LED = Light Emitting Diode; converts electrical energy to light; more efficient than bulb |
| 6 | Solar cell = photovoltaic diode; converts light to electricity |
| 7 | Transistor invented = **1947** (Bell Labs; Bardeen, Brattain, Shockley); Nobel 1956 |
| 8 | Integrated Circuit (IC) invented = **1958** (Jack Kilby at TI; Nobel 2000) |
| 9 | IC generations: SSI, MSI, LSI, VLSI, ULSI (Ultra-large scale integration) |
| 10 | Moore's Law: transistor count on chip doubles every ~2 years (Gordon Moore, 1965) |

| # | Fast Fact |
|---|---|
| 11 | 1 byte = 8 bits; 1 KB = 1024 bytes; 1 MB = 1024 KB; 1 GB = 1024 MB |
| 12 | Data sizes: bit < nibble(4 bits) < byte(8) < KB < MB < GB < TB < PB < EB |
| 13 | Binary: 0,1; Octal: 0-7 (base 8); Decimal: 0-9 (base 10); Hexadecimal: 0-9,A-F (base 16) |
| 14 | CPU = ALU (Arithmetic Logic Unit) + CU (Control Unit) + Registers |
| 15 | RAM = volatile (lost on power off); ROM = non-volatile (permanent) |
| 16 | Cache memory = fastest memory; SRAM; between CPU and RAM |
| 17 | First computer (ENIAC) = 1945; used vacuum tubes; weighed 30 tons |
| 18 | 1st generation = vacuum tubes; 2nd = transistors; 3rd = ICs; 4th = microprocessors; 5th = AI |
| 19 | Microprocessor invented = **1971** (Intel 4004; Federico Faggin) |
| 20 | Internet began as **ARPANET** (1969, USA; Defense department) |
| 21 | WWW (World Wide Web) invented = **Tim Berners-Lee (1989)**; CERN, Switzerland |
| 22 | .com domain = commercial; .org = organization; .gov = government; .edu = education |
| 23 | GPS = Global Positioning System; 24+ satellites; US DoD; free civilian use |
| 24 | Nanotechnology = 1-100 nm scale; applications: drug delivery, sensors, materials |
| 25 | Quantum bits (qubits) = can be 0, 1, or both simultaneously (superposition) |
| 26 | 5G frequency bands = sub-6 GHz + mmWave (above 24 GHz) |
| 27 | Optical fibre internet = uses light pulses through silica glass; faster than copper |
| 28 | Blockchain = distributed ledger; each block cryptographically linked; basis of cryptocurrency |
| 29 | Deep Blue (IBM, 1997) = first chess computer to defeat world champion (Kasparov) |
| 30 | ChatGPT (OpenAI, 2022) = large language model AI; generative AI milestone |